

DUSTBADGE · DESIGN DECISIONS REGISTER

DustBadge design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	DBG-DEC-001	Version	0.2
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dustbadge		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

DustBadge design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

All are proposed, awaiting Amish.

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Review the design-for-construction changes P1 to P10 (shell screws and tubes, gasket, port seals, screen, stop ribs, module placement, board size, cell ribs, clip screws, light pipe)	Accept; or change any of them	Accept: they keep what the badge does, its size and its cost	The whole build plan	DBG-DDR-003, Table 1
2	Cell swelling allowance: 0.7 mm of free depth (about 6 % of the cell's thickness) in the lithium-cell safety case	(a) accept for the prototype, inspect at each charge and retire at the first swelling; (b) deepen the badge 1 mm and add a foam pad (about 1.5 g more, R9 no longer met)	(a), revisited at TRL 4 with the bought cell's datasheet	Safety stops S1 and S7; rear shell depth	DBG-DDR-003, A1
3	On and off: the concept has no switch	(a) firmware only: the boost switched off through its enable input, the controller asleep, waking on USB power; (b) a reed switch inside with a magnet outside; (c) a sealed push button (a new opening)	(a): no change to the shell	Wiring (the enable line is already wired)	DBG-DDR-003, A2
4	Mass margin for R9: 0.1 g on catalogue masses	(a) accept, weigh the prototype at TRL 4; (b) look for mass now, for example a 1.8 mm rear wall	(a)	First check "mass and size"	DBG-DDR-003, A3; DBG-CAL-001 [J2]
5	First co-design partner (worker organization, NGO or university hygiene group)	Partner to be named	None given	Not part of the TRL 3 build; needed for field trials	DBG-DDR-001, O1
6	How a site silica fraction is obtained where no laboratory is near	Partner laboratory, field infrared method, or other	None given	Not part of the TRL 3 build; needed before silica estimates are shown	DBG-DDR-001, O2
7	Clear front window over the sensor, shown in the product renders only	(a) renders only, decide at TRL 4; (b) adopt, adding a sealed joint that bears on R10	(a)	None now; front shell if adopted	REVIEW 2026-09-26, item 1
8	Visible sensor fan in the renders (illustrative; the real sensor encloses its fan)	Keep and caption as illustrative; or remove	Keep, captioned	None	REVIEW 2026-09-26, item 2
9			Recess them if adopted		

#	Decision needed	Options	Recommendation	Affects in the build	Source
	Side grip ribs in the renders stand 0.6 mm proud (65.2 mm wide)	Accept for appearance; or recess them to keep 64 mm		Front shell, if adopted	REVIEW 2026-09-26, item 4
10	Lanyard with a breakaway buckle, shown in the renders, has no BOM line	Add an optional line (about \$1) at TRL 4; or leave out	Add at TRL 4; it would take the estimate about \$1 over the value-engineering target	None now	REVIEW 2026-09-26, item 5

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The cell, including its protection board, is no larger than 50 x 34 x 11.5 mm, its lead leaves at one short side, and its plug polarity matches the charger	The cell ribs, the 0.7 mm swelling allowance and the board notch are sized for it	DBG-DDR-003, P8
2	The charger module is no larger than 11 x 19 mm with its USB-C on a short edge, and its temperature window is 0 to 45 °C	It must fit beside the sensor with its USB-C in the opening, and the charge stop is part of the safety case	DBG-DDR-003, P6
3	The boost module (11.4 x 9 mm or less, with an enable input) and the humidity breakout (10 x 10 mm or less, sensor at one edge)	They are taped into a strip 17 mm wide beside the sensor; the humidity sensor must sit over its vent	DBG-DDR-003, P6
4	Where the particle sensor's cable leaves it, and the cable length	The model leaves room above and beside the sensor between the stop ribs; a connector on another face would need the ribs moved	DBG-DDR-003, P5
5	The spring clip's base leaf is flat and wide enough for two holes 12 mm apart	The clip is held by two M2 x 8 screws	DBG-DDR-003, P9
6	The pilot hole for the M2 thread-forming screws in PETG (1.7 mm assumed), from the screw maker's data	Too large a hole strips; too small cracks the boss	DBG-DDR-003, P1

■ Value engineering

Value-engineering target: USD 91 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 91.00 (USD 0.00 over or under the target, so on the target with no margin). The estimate rests on indicative prices.

Main cost drivers: the optical particle sensor (USD 48.00, 53 % of the cost), the 2,000 mAh LiPo cell (USD 13.00; it added USD 3), the controller and BLE module (USD 10.00) and the carrier board (USD 9.00).

Savings worth trying: confirming prices when parts are bought. A 1,500 mAh cell would save USD 3 but returns R7 (a full shift) to at risk. The optional lanyard (open decision 10) would take the estimate about USD 1 over the target.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D7: SPS30-class sensor, silica shown only as a labeled estimate, US OSHA and MSHA default limits, continuous sampling, worker-owned data, surface quarries and stone fabrication first, downward screened inlet with a yellow shell and clip	Amish: "i accept all your recommendations, go with them across all repos."	DBG- DDR-001, DBG- DDR-002
2026-09-25	D8 to D11: 2,000 mAh cell; R3 reference rule (4.2 L/min cyclone over two shifts at low-dust, high-silica sites); sun use rule; R2 restated to 1 mg/m ³ with an over-range flag	Amish, same instruction	DBG- DDR-002
2026-09-26	Value-engineering target set to \$91 to match the priced BOM (O3)	Amish: "i approve all the budget items."	DBG- DDR-002 v0.2